A MORPHISM IS QUASI-COMPACT IF ITS COMPOSITE WITH A QUASI-SEPARATED MORPHISM IS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is short but complete: each of the two factors of f is shown quasi-compact and the composition statement closes the argument.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- A section of a quasi-separated morphism is quasi-compact. Lemma lemma-section-immersion, used at proof lines 1462-1465. Makes the graph (1, f) quasi-compact.
- Stability of quasi-separatedness under base change. Lemma lemma-base-change-separated, used at proof lines 1464-1465. Identifies X x_Z Y -> X as a base change of g.
- Stability of quasi-compactness under base change. Lemma lemma-base-change-quasi-compact, used at proof lines 1466-1468. Makes the projection X x_Z Y -> Y quasi-compact.
- A composite of quasi-compact morphisms is quasi-compact. Lemma lemma-composition-quasi-compact, used at proof lines 1469-1470. Concludes for f.

Source reference. Original lemma 03HF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4565–4565. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4681–4681. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4566–4566. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4685–4685. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of algebraic spaces over S. If $g \circ f$ is quasi-compact and g is quasi-separated then f is quasi-compact.*

Proof. This is true because f equals the composition $(1, f) : X \to X \times_Z Y \to Y$. The first map is quasi-compact by Lemma 03KP because it is a section of the quasi-separated morphism $X \times_Z Y \to X$ (a base change of g, see Lemma 03KL). The second map is quasi-compact as it is the base change of $g \circ f$, see Lemma 03HF. And compositions of quasi-compact morphisms are quasi-compact, see Lemma 03HG. □